Lynea Database Schema

Purpose: This document defines the proposed database tables, columns, data types, and key constraints for Lynea, an AI-assisted group travel planning application. This is an early-stage schema for team review; it can be adjusted as product requirements become clearer.

1. users

Column	Data type	Key / constraint	Description
user_id	UUID	PK	Unique identifier for a user
name	VARCHAR(100)	NOT NULL	Display name
email	VARCHAR(255)	UNIQUE, NOT NULL	Login/contact email
created_at	TIMESTAMP WITH TIME ZONE	NOT NULL; default now()	Account creation time

2. trip_intents

Column	Data type	Key / constraint	Description
intent_id	UUID	PK	Unique travel request
user_id	UUID	FK → users.user_id; NOT NULL	Person who submitted the request
budget_min	DECIMAL(10,2)	NULL allowed	Minimum estimated budget per person
budget_max	DECIMAL(10,2)	NULL allowed	Maximum estimated budget per person
earliest_start_date	DATE	NULL allowed	Earliest acceptable departure
latest_end_date	DATE	NULL allowed	Latest acceptable return
destination_preference	TEXT	NULL allowed	Destination, region, or flexible/anywhere
travel_style	VARCHAR(100)	NULL allowed	Broad style, such as adventure, beach, or city
status	VARCHAR(20)	NOT NULL	Suggested values: open, matched, withdrawn
created_at	TIMESTAMP WITH TIME ZONE	NOT NULL; default now()	Request creation time

3. travel_groups

Column	Data type	Key / constraint	Description
group_id	UUID	PK	Unique travel group

group_name	VARCHAR(150)	NULL allowed	Group display name
status	VARCHAR(20)	NOT NULL	Suggested values: forming, planning, finalized, cancelled
created_at	TIMESTAMP WITH TIME ZONE	NOT NULL; default now()	Group creation time

4. group_members

Column	Data type	Key / constraint	Description
group_member_id	UUID	PK	Unique membership record
group_id	UUID	FK → travel_groups.group_id; NOT NULL	Group joined or invited to
user_id	UUID	FK → users.user_id; NOT NULL	Member user
joined_at	TIMESTAMP WITH TIME ZONE	NULL allowed	When membership was accepted
status	VARCHAR(20)	NOT NULL	Suggested values: invited, accepted, declined, left
role	VARCHAR(20)	NOT NULL	Suggested values: member, organizer
constraint	UNIQUE (group_id, user_id)	Unique membership	Prevents the same user appearing twice in one group

5. trip_preferences

Column	Data type	Key / constraint	Description
preference_id	UUID	PK	Unique preference record
group_member_id	UUID	FK → group_members.group_member_id; UNIQUE, NOT NULL	Member whose preferences these are; one record per membership initially
budget_min	DECIMAL(10,2)	NULL allowed	Minimum preferred budget per person
budget_max	DECIMAL(10,2)	NULL allowed	Maximum preferred budget per person
start_date	DATE	NULL allowed	Preferred trip start date
end_date	DATE	NULL allowed	Preferred trip end date
destination_preference	TEXT	NULL allowed	Preferred destination or region
travel_style	VARCHAR(100)	NULL allowed	General travel style
notes	TEXT	NULL allowed	Additional relevant preferences

updated_at	TIMESTAMP WITH TIME ZONE	NOT NULL; default now()	Last update time

6. interests

Column	Data type	Key / constraint	Description
interest_id	UUID	PK	Unique interest identifier
interest_name	VARCHAR(80)	UNIQUE, NOT NULL	Standardized interest, e.g., hiking, food, nightlife

7. member_interests

Column	Data type	Key / constraint	Description
preference_id	UUID	FK → trip_preferences.preference_id; part of composite PK	Preference record
interest_id	UUID	FK → interests.interest_id; part of composite PK	Selected interest
priority	INTEGER	NOT NULL; suggested range 1–5	Importance of this interest to the person
composite primary key	(preference_id, interest_id)	PK	Prevents duplicate interest selections for one preference record

8. recommendation_runs

Column	Data type	Key / constraint	Description
run_id	UUID	PK	Unique AI recommendation run
group_id	UUID	FK → travel_groups.group_id; NOT NULL	Group receiving recommendations
status	VARCHAR(20)	NOT NULL	Suggested values: pending, completed, failed
model_version	VARCHAR(100)	NULL allowed	Model or agent version, if useful
created_at	TIMESTAMP WITH TIME ZONE	NOT NULL; default now()	When generation started
completed_at	TIMESTAMP WITH TIME ZONE	NULL allowed	When generation completed

9. trip_options

Column	Data type	Key / constraint	Description
option_id	UUID	PK	Unique generated trip option
run_id	UUID	FK → recommendation_runs.run_id; NOT NULL	AI run that produced this option
destination_name	VARCHAR(200)	NOT NULL	Suggested destination
estimated_cost_per_person	DECIMAL(10,2)	NULL allowed	Estimated cost per traveler
description	TEXT	NULL allowed	Summary of the trip idea
match_score	DECIMAL(5,2)	NULL allowed	Optional group match score; define scale consistently
match_explanation	TEXT	NULL allowed	Why the AI recommended the option and its trade-offs
created_at	TIMESTAMP WITH TIME ZONE	NOT NULL; default now()	When option was saved

10. votes

Column	Data type	Key / constraint	Description
vote_id	UUID	PK	Unique vote record
option_id	UUID	FK → trip_options.option_id; NOT NULL	Trip option being rated
group_member_id	UUID	FK → group_members.group_member_id; NOT NULL	Member submitting the vote
vote_value	VARCHAR(20)	NOT NULL	Suggested values: like, maybe, dislike
created_at	TIMESTAMP WITH TIME ZONE	NOT NULL; default now()	Vote time
constraint	UNIQUE (option_id, group_member_id)	Unique vote per member per option	Update an existing vote if a member changes their choice